

My interest in machine learning began with a question I couldn't stop thinking about: could the applications I was building learn from their users, rather than just serve them? That question pulled me from software development into ML, and it has shaped every project I've worked on since.

My formal entry point was a two-day workshop at my college where I encountered supervised learning and model evaluation for the first time. Rather than stopping there, I continued independently working through online resources, experimenting with datasets, and completing Google's 5-Day AI Agents Intensive Course to understand how modern intelligent systems are structured. The most important shift came when I stopped reading about ML and started building with it. Theoretical concepts only became real when I had to make actual decisions which features to keep, which model to trust, and why a metric mattered in a specific context.

My most significant project is a Job Application Screening System — an end-to-end pipeline that predicts candidate suitability from profile data. I built the full pipeline from data cleaning, feature encoding to train-test split, model training, evaluation and compared a Linear Regression baseline against a Random Forest Regressor to capture non-linear relationships between candidate attributes. The Random Forest model outperformed the baseline significantly, and I deployed it via Gradio so non-technical users could interact with it directly. The most instructive moment wasn't the accuracy gain; it was when early predictions were skewed by improperly handled categorical variables. That mistake forced me to rebuild the preprocessing stage entirely and taught me that understanding your data matters more than choosing your model.

My current gaps are concrete. I understand how to train and evaluate models on clean, bounded datasets but I have no exposure to how systems behave under distribution shift, how retraining is scheduled in production, or how recommendation models balance relevance with latency at scale. These aren't gaps I can close through another course. They require reasoning that only comes from practitioners who've debugged these problems in real deployments. Amazon ML Summer School offers exactly the perspective of engineers who build ML systems serving millions of users, in areas like supervised learning theory, large-scale ML, and recommendation systems where Amazon has built some of the most influential real-world implementations.

My background sits at the intersection of Computer Science and Design, which means I approach model building with both implementation instincts and user-impact awareness. I think about what a prediction does to the person receiving it, not just whether the metric is good. Combined with a 9.18 CGPA reflecting consistent academic depth and a self-directed learning habit that has taken me well beyond coursework, I bring both the foundation and the hunger to make the most of this program.

My immediate next step after MLSS is implementing embedding-based semantic matching in SkillSwap, my peer-to-peer skill exchange platform, replacing the filter logic it currently uses. Beyond that, I want to build ML systems where the model is central to the product, not an afterthought. Amazon ML Summer School is the clearest path I've found toward that kind of depth.